The local Plectic Conjecture

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(joint work with Tony Feng, Matteo Tamiozzo in progress)

The Plectic Conjecture is raised by Jan Nekovář and Anthony Scholl [1], which predicts that the cohomology of Shimura varieties attached to a reductive group arising from Weil restriction of scalars has a larger symmetry (the so-called plectic structures). We discuss in this talk the local version of the conjecture and a proof of the local conjecture for abelian type Shimura varieties using the machinery of Fargues-Scholze local Langlands correspondence [2] and the Igusa stacks [3].

More precisely, let F be a totally real field of degree d and H/F be a reductive group. Let $\mathsf{G} := \mathrm{Res}^{\mathsf{F}}_{\mathbb{Q}} \mathsf{H}$ be the Weil restriction of scalars of H from F to $\mathbb{Q}$. Assume it is part of a Shimura datum (G, X) with Hodge cocharacter μ and reflex field E. Then the natural action of the Galois group $\Gamma_{\mathbb{Q}}$ on the cocharacter group $X_*(\mathsf{G})$ lifts naturally to the "plectic Galois group" $\Gamma^{\mathrm{plec}}_{\mathsf{F}/\mathbb{Q}} := \mathrm{Aut}_{\mathsf{F}}(\mathsf{F} \otimes \overline{\mathbb{Q}})$. We may define the plectic reflex Galois group $\Gamma^{[\mu]}_{\mathsf{F}/\mathbb{Q}}$ to be the stabilizer of the $\mathsf{G}(\overline{\mathbb{Q}})$-conjugacy class $[\mu]$ in $\Gamma^{\mathrm{plec}}_{\mathsf{F}/\mathbb{Q}}$.

Consider the Shimura variety attached to (G, X), i.e. a tower of algebraic varieties $\{\mathbf{Sh}_K\}_K$ over E for K running through neat compact open subgroups of $\mathsf{G}(\mathbb{A}_f)$. Let ℓ be a prime number and we fix an isomorphism $\mathbb{C} \simeq \overline{\mathbb{Q}}_\ell$. For an algebraic representation ξ of G with $\overline{\mathbb{Q}}_\ell$-coefficient, we denote by $\mathcal{L}_\xi$ the étale $\overline{\mathbb{Q}}_\ell$-local system attached to it. Write $R\Gamma(\mathbf{Sh}_\infty, \mathcal{L}_\xi)$ for the colimit of the cohomology groups $R\Gamma_{\text{ét}}(\mathbf{Sh}_{K,\overline{\mathbb{Q}}}, \mathcal{L}_\xi)$. The (global) Plectic Conjecture can be formulated as follows:

Conjecture 1 (Nekovář-Scholl). *The complex $R\Gamma(\mathbf{Sh}_\infty, \mathcal{L}_\xi)$ in $D^b(\Gamma_{\mathsf{E}}, \overline{\mathbb{Q}}_\ell)$ lifts canonically to an object in $D^b(\Gamma^{[\mu]}_{\mathsf{F}/\mathbb{Q}}, \overline{\mathbb{Q}}_\ell)$, compatibly with the Hecke actions.*

This conjecture is partly motivated by its application to semi-simplicity of the cohomology of Shimura varieties as Galois representations, see [4], [5]. It is still widely open, except for results for CM points [6], [7] and partial results for quaternionic Shimura varieties and Hilbert modular varieties [4], [1]. Other related results include the construction of partial Frobenii as predicted by the conjecture in the case of abelian type Shimura varieties [8], [9], the local version of the conjecture (see below) on the basic local of abelian type Shimura varieties [10], and a Lie algebra version for Hilbert modular varieties [11].

To state the local version of the conjecture, we fix a rational prime p and an isomorphism $\iota : \mathbb{C} \simeq \overline{\mathbb{Q}}_p$, this induces a p-adic place ν of E, we denote by E the completion E_ν. Write F_p for $\mathsf{F} \otimes_{\mathbb{Q}} \mathbb{Q}_p$ and we can similarly define the local plectic reflex Galois group $\Gamma^{[\mu]}_{\mathsf{F}_p/\mathbb{Q}_p}$ to be the stabilizer of $[\mu]$ (viewed as a cocharacter of $G := \mathsf{G}_{\mathbb{Q}_p}$ via ι) in $\mathrm{Aut}_{\mathsf{F}_p}(\mathsf{F} \otimes_{\mathbb{Q}} \mathbb{Q}_p)$. Restricting the cohomology of the Shimura varieties to the local Galois groups, we have the following local Plectic Conjecture.

Conjecture 2 (The local Plectic Conjecture). *The complex $R\Gamma(\mathbf{Sh}_\infty, \mathcal{L}_\xi)$ in $D^b(\Gamma_E, \overline{\mathbb{Q}}_\ell)$ lifts canonically to an object in $D^b(\Gamma^{[\mu]}_{\mathsf{F}_p/\mathbb{Q}_p}, \overline{\mathbb{Q}}_\ell)$, compatibly with the global conjecture via restriction along the natural map $\Gamma^{[\mu]}_{\mathsf{F}_p/\mathbb{Q}_p} \to \Gamma^{[\mu]}_{\mathsf{F}/\mathbb{Q}}$. In particular, when p splits completely in F, there exist partial Frobenii at p via the identification $\Gamma^{[\mu]}_{\mathsf{F}_p/\mathbb{Q}_p} \simeq \prod_{i=1}^d \Gamma^{[\mu_i]}_{\mathbb{Q}_p}$.*

The main result of our work in progress [12] is follows:

Theorem 3 (In progress). *For $p \neq \ell$ and abelian type Shimura data, the action of the local plectic reflex Galois group (in fact a Weil group version) on the cohomology of the corresponding Shimura varieties exists canonically.*

Let us explain briefly the idea of the proof. For simplicity of the formalism, we fix a torsion $\mathbb{Z}_\ell$-algebra Λ as the coefficient ring, and consider only the cohomology of Shimura varieties with constant coefficient. (The result for $\overline{\mathbb{Q}}_\ell$ can be obtained by taking an inverse limit of such and base-change.) Consider the v-stack Bun_G (over $\overline{\mathbb{F}}_p$) of G-bundles on the Fargues-Fontaine curve as in [2] and $D(\mathrm{Bun}_G)$, the derived category of étale sheaves on Bun_G with Λ-coefficients. Fargues–Scholze constructed Hecke operators, acting as endo-functors on $D(\mathrm{Bun}_G)$, whose essential image carry internally an action of the Weil group $W_{\mathbb{Q}_p}$ (or a subgroup of it). The main result (existence of abelian type Igusa stacks) of [3] leads to a formula of the cohomology of abelian type Shimura varieties in terms of these operators. Namely, we have the following:

Theorem 4 (In progress). *Let $p \neq \ell$ and (G, X) be an abelian type Shimura datum. Then there is a complex $\mathcal{F} \in D(\mathrm{Bun}_G)$, such that*

$$R\Gamma(\mathbf{Sh}_\infty, \Lambda) = i_1^* T_\mu(\mathcal{F})$$

as $G(\mathbb{Q}_p) \times W_E$-representations. Here T_μ is a Hecke operator attached to the tilting representation of the Langlands dual group $\widehat{G}$ labelled by μ, and $i_1 : \underline{BG(\mathbb{Q}_p)} \hookrightarrow \mathrm{Bun}_G$ is the inclusion of the open stratum corresponding to the trivial G-bundle.

Now the idea is straight-forward: Upon reconciling the way Fargues–Scholze constructed the Hecke operators (using a Beilinson-Drinfeld affine Grassmannian with the sheaf of degree-one Cartier divisors Div^1 as base), which made the action of the Weil group appear, we could construct a plectic version T^{plec}_μ of the Hecke operator (using a Beilinson-Drinfeld affine Grassmannian with a modification of Div^1 as base). Objects in its essential image carry internally a plectic reflex Weil group action. In fact, this part has already been worked out by Li-Huerta in [10], so we do not claim originality. The desired result then follows from the compatibility of the usual and plectic Hecke operators and an analogue of Theorem 4, replacing T_μ by T^{plec}_μ. It is also worth mentioning that with the abelian type Igusa stack, one can construct a plectic companion of the Shimura variety, whose cohomology realizes the lift of $R\Gamma(\mathbf{Sh}_\infty, \Lambda)$ to a complex with local plectic reflex Galois group action.

References

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